



VEHICLE SERVICE CENTRE MANAGEMENT SYSTEM

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A centralized database for every step of a service visit

When a customer brings a vehicle for servicing, a lot of information needs to be tracked. This system organizes all of it in one place:



Customer Details



Vehicle Details



Service Booking



Problems Reported



Job Card



Mechanic Assignment



Repairs & Tasks



Spare Parts Used



Invoice



Payment



Previous Service History

THE PROBLEM

Paper records don't scale with a busy service centre

Many service centres still rely on paper job cards, registers or scattered records — creating problems such as:



Difficulty tracking vehicle service history



Difficulty managing spare-part stock



Loss or duplication of records



Billing errors



Difficulty tracking ongoing jobs



Difficulty generating reports

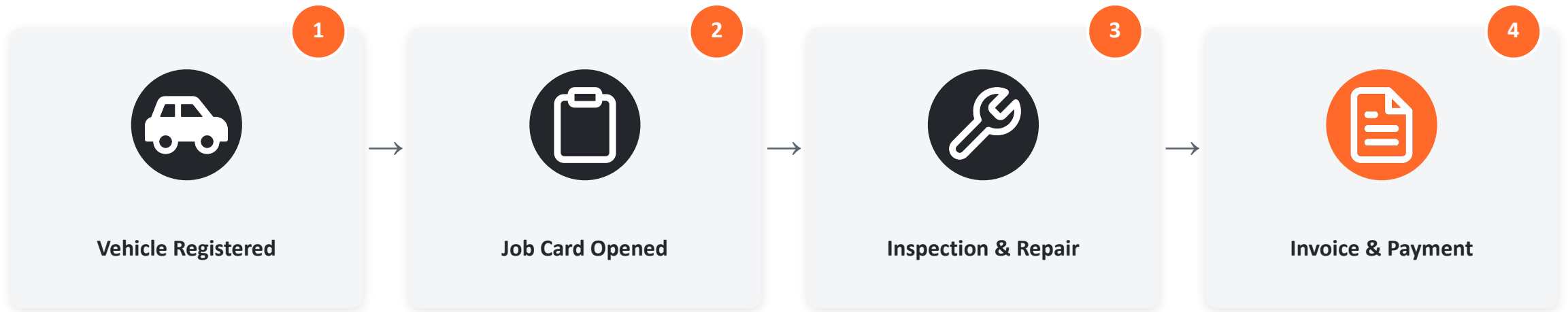


Improper management of mechanic assignments

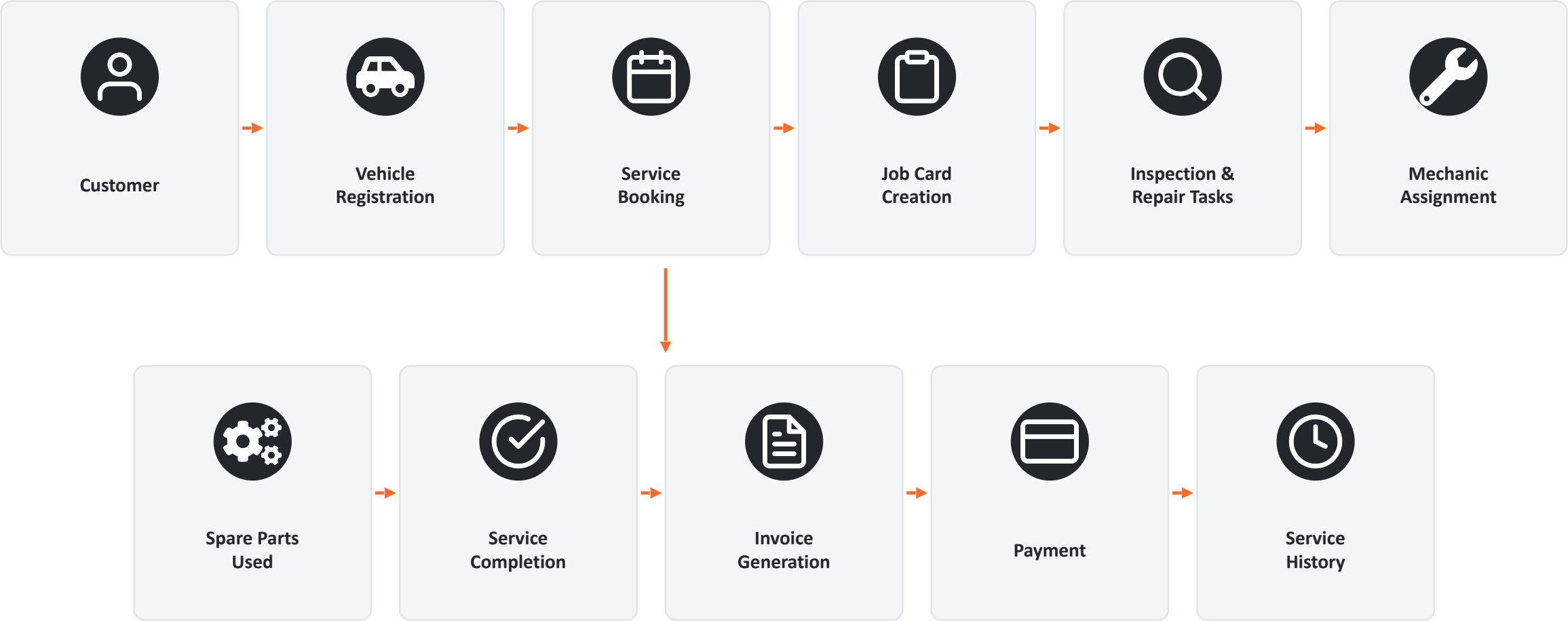
→ We need a centralized database system to manage all service-centre information efficiently.

The complete journey of a vehicle through the centre

The process starts when a customer registers their vehicle and makes a service booking. **When the vehicle arrives, a job card is created** to record inspection details, repair tasks, assigned mechanics, spare parts used and service status. Once service is complete, an **invoice is generated and payment is recorded**, and the system maintains the vehicle's complete service history.



How the system works



PROJECT ROADMAP

How we plan to develop the project

STAGE 1

Requirement Analysis

Understand the requirements of a vehicle service centre and identify the information to be stored.



STAGE 2

Database Design

Identify main entities (Customer, Vehicle, Booking, Job Card, Mechanic, Spare Part, Invoice, Payment) and their relationships; build the ER diagram.



STAGE 3

Relational Database Design

Convert the ER diagram into database tables with primary keys, foreign keys and constraints.



STAGE 4

Database Implementation

Implement the database using MySQL; create tables and populate them with realistic sample data.



STAGE 5

SQL Queries & Reports

Write SQL queries for pending jobs, service history, mechanic workload, spare-part usage and revenue.



STAGE 6

Front-End Development

Build a simple front-end application using Java or Python to interact with the database.



EXPECTED OUTCOME

What the finished system delivers



Reduce manual paperwork



Maintain organized records



Track service progress



Manage mechanics



Track spare parts



Generate accurate invoices



Maintain vehicle service history



Generate useful reports

CONCLUSION

A centralized system for the complete service journey

Our project manages the complete service process — from customer and vehicle registration, through booking, servicing, mechanic assignment and spare-part usage, to billing and payment.

NEXT STEPS

- Design the ER model
- Create the relational schema
- Implement the database using MySQL

Thank You